

HARI CHARAN V

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ABOUT

An Electrical and Electronics Engineering student passionate about embedded systems, renewable energy, and power electronics application development. Strongly driven by AI integration and leveraging generative AI tools to optimize electrical systems.

SKILLS

- MATLAB/Simulink, LTspice, Keil Micro-Vision
- Python, Embedded C
- PCB Design (KiCad, EasyEDA)
- Internet of Things (IoT)
- Embedded Systems
- AI Application in Electrical Systems

EXPERIENCE

Intern | MicroLabs, Chennai 06/2026

- Evaluated and tested various electrical and electronic sensors for automotive performance and endurance.
- Developed and deployed a custom mobile application for real-time tracking of orders and samples.

Intern | Center for Smart Grid, VIT Chennai 05/2025 - 07/2025

- Designed a hybrid solar-wind-fuel cell power generation system in MATLAB/Simulink.
- Implemented a feed-forward Artificial Neural Network (ANN) to optimize solar PV MPPT efficiency.

PROJECTS

SeizureGuard: Multi-Sensor Wearable Bio-Telemetry PCB

- Routed a compact health-monitoring PCB with optimized I2C bus tracking and analog skin conductance processing.

AI-ML Driven State Estimation for Smart Grid Applications

- Implemented an ANN-based power system state estimation model on the IEEE 14-bus system using MATPOWER.

Real-Time Power Monitoring and Power Factor Correction System

- Built a real-time monitoring system using a PZEM module and relay-based capacitor switching to improve power factor.

Adaptive Headlight Intensity Control System using Microcontroller

- Developed an LDR-based adaptive automotive headlight control loop implemented on an RC car platform.

EDUCATION

B.Tech: Electrical and Electronics Engineering Expected 04/2027

Vellore Institute of Technology, Chennai

LANGUAGES

English, Hindi, Tamil, Kannada

CERTIFICATIONS

CSIR-CEERI Chennai Centre – Embedded Control for Power Converters (May – July 2026)

Cognitive Class (IBM Developer Skills Network) – SQL and Relational Databases 101 (June 9, 2026)

Forage (Tata) – ESG Job Simulation (June 8, 2026)